

Reflect-R1: Evidence-Driven Reflection for Self-Correction in Long Video Understanding

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Project: <https://github.com/ShuimuChen-hyq/Reflect-R1>

Abstract. Current multimodal reflection mechanisms for long video understanding predominantly rely on closed-loop self-reflection within internal parameters. Lacking objective external evidence, models are frequently trapped in blind confidence and often fail to correct errors. Furthermore, applying reinforcement learning to multi-stage reflection pipelines introduces severe policy coupling, which is exacerbated by a critical scarcity of dedicated training data. To address these limitations, this work proposes Reflect-R1, the first Evidence-Driven self-correction framework for long video understanding. The framework constructs a three-stage pipeline consisting of intuition, verification, and arbitration. By dynamically retrieving objective visual evidence to verify initial intuitions and autonomously executing multiple temporal searches to resolve conflicts, it completely breaks the hallucination loop. To overcome policy coupling, we design a stage-decoupled reinforcement learning algorithm named SD-GRPO that independently computes advantage functions across different reasoning stages. Concurrently, we construct a dataset of 120K samples to bridge the training data gap. Extensive experiments on benchmarks such as VideoMME and LongVideoBench demonstrate that Reflect-R1 achieves state-of-the-art performance. Our method significantly improves the genuine rectification rate and enables authentic self-correction strictly grounded in objective evidence.

Keywords: Reinforcement Learning · Self-reflection · Multimodal Large Model · Long Video Understanding

1 Introduction

Long video understanding [5,9,11,25,30,40] is a critical task for applying artificial intelligence to complex real-world scenarios. Recent video-oriented multimodal

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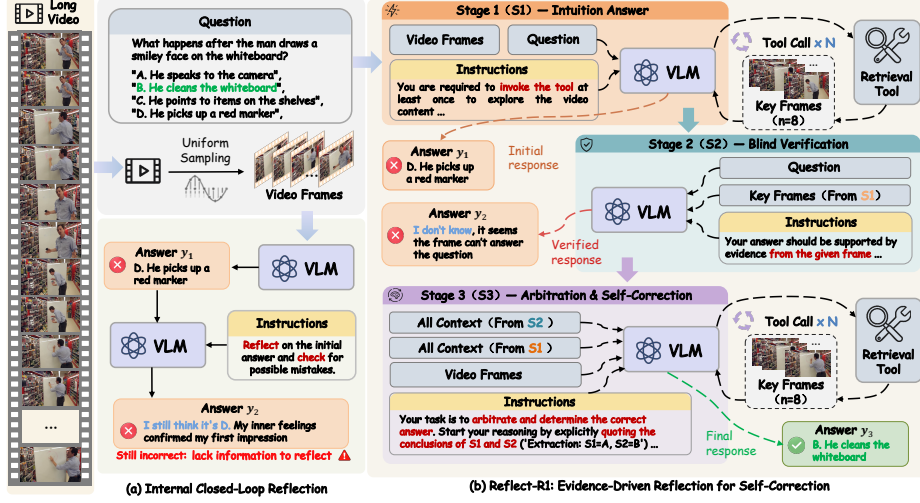


Fig. 1: Comparison between Internal Closed-Loop Reflection and Evidence-Driven Reflection. (a) Traditional closed-loop reflection relies solely on internal parametric knowledge, easily falling into the trap of blind confidence and failing to correct errors. (b) Reflect-R1 completely breaks the hallucination loop by formalizing an “intuition-verification-arbitration” pipeline, executing active searches to achieve genuine self-correction strictly grounded in objective retrieved evidence.

large language models (MLLMs) further broaden this landscape to event streams, personalized video chat, and fine-grained facial video understanding [17, 28, 47]. Beyond understanding visual content, reliable deployment also requires reflection, where a model explicitly scrutinizes and potentially corrects its initial intuitive output before generating a final response. Recent pioneering works, including Vision-R1 [12], VL-Rethinker [33], and Video-R1 [6], have advanced the field by integrating such reflection mechanisms into MLLMs to mitigate hallucinations and capture visual details. Concurrently, large language models such as OpenAI o1 [14] and DeepSeek-R1 [10] demonstrate that eliciting long chain-of-thought (CoT) reasoning [15, 37, 45] through reinforcement learning substantially enhances reflection and complex logical deduction. However, bringing such reflection paradigms to multimodal long-video understanding is difficult. Most existing methods perform reflection in a closed-loop internal manner, which gives rise to two main problems.

The first problem is verification failure caused by a lack of objective evidence. As illustrated in Fig. 1 (a), this paradigm forces the model to rely solely on internal knowledge to repeatedly scrutinize the initial answer (y_1). Because long video information is complex and independent external visual evidence is absent, this completely closed internal reasoning process easily traps the model in blind confidence [16, 23]. When attempting to correct errors, the model frequently uses internally generated hallucinations to forcibly justify the initial erroneous conjecture, making the reflection process ineffective or even counterproductive. The empirical analysis in Sec. 2 clearly confirms this phenomenon. Without external

verification, the reflection process of multimodal models often degenerates into random alterations [23], where the probability of changing a correct answer to an incorrect one frequently exceeds the probability of correcting errors.

The second problem involves policy coupling [34] during reinforcement learning optimization and an acute scarcity of training data. To elicit reflection capabilities through reinforcement learning, it is intuitively necessary to jointly train the initial answering phase and the subsequent correction phase within a unified trajectory. However, applying standard reinforcement learning directly to such a long-chain, multi-stage process triggers severe policy coupling. Specifically, the complexity of prolonged reasoning drives the model to exploit optimization shortcuts, such as simply repeating the initial guess during the reflection stage to secure base rewards instead of learning authentic error-correction logic [4]. Furthermore, the extreme lack of high-quality training data tailored for multimodal reflection remains a critical bottleneck that prevents models from developing deep self-correction abilities.

To address verification failures caused by closed-loop blind confidence, we propose Reflect-R1. To the best of our knowledge, Reflect-R1 is the first evidence-driven self-correction framework for long video understanding that explicitly decomposes reflection into an intuition-verification-arbitration pipeline. In the first stage, the model generates an intuitive answer (y_1) and actively retrieves keyframes as external visual evidence. In the second stage, the model performs an independent blind verification (y_2) by relying exclusively on these retrieved frames to assess the initial intuition. This process ensures that the model evaluates the question based on objective evidence while maintaining strict information isolation from the global video context. Finally, the arbitration stage (y_3) resolves conflicts between the subjective intuition and the objective verification result to produce a final response. If the initial evidence is insufficient for a definitive conclusion, the model autonomously re-invokes temporal search tools until conclusive proof is captured. By grounding the entire reflection process in external visual evidence, Reflect-R1 effectively breaks the hallucination loop and achieves authentic multimodal self-correction.

Furthermore, to overcome policy coupling in GRPO [27] and bridge the training data gap, we design a novel Stage-Decoupled GRPO (SD-GRPO) algorithm along with dedicated datasets. The SD-GRPO algorithm effectively prevents the model from seeking optimization shortcuts by computing the advantage function independently across different reasoning stages, including intuition, verification, and arbitration. This mechanism forces the model to learn genuine error correction logic. Concurrently, we systematically construct Reflect-R1-CoT-90k for supervised fine-tuning cold start and Reflect-R1-RL-30k for reinforcement learning training, fully resolving the data scarcity bottleneck in this field.

Reflect-R1 achieves state-of-the-art performance across major benchmarks including VideoMME [7], LongVideoBench [40], and MLVU [48]. It also demonstrates superior localization precision in temporal search tasks on Haystack-LVBench [43]. Most importantly, Reflect-R1 exhibits exceptional reflection reliability. While existing internal reflection paradigms frequently suffer from perfor-

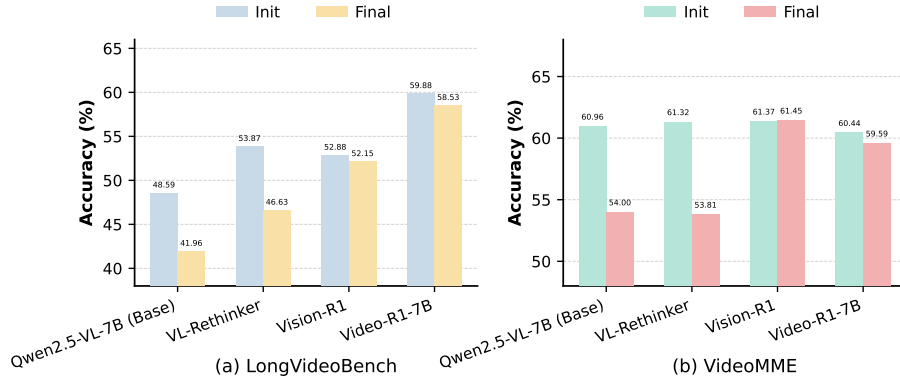


Fig. 2: The Failure of Internal Reflection. Without objective external visual anchors, static closed-loop reflection tends to amplify initial visual hallucinations.

mance degradation, our framework achieves consistent accuracy improvements of +2.82% on LongVideoBench and +1.41% on VideoMME. These results confirm that grounding self-correction in objective evidence effectively breaks the hallucination loop in long video understanding.

In summary, our main contributions are as follows:

- We propose Reflect-R1, the first Evidence-Driven self-correction framework. It effectively mitigates reflection failures caused by the lack of external evidence, enabling authentic self-correction grounded in objective clues.
- We design SD-GRPO, a stage-decoupled reinforcement learning algorithm that independently computes advantage functions to overcome policy coupling in multi-stage reasoning. Additionally, we construct a dedicated dataset of 120K samples to bridge the training data gap for multimodal reflection.
- Reflect-R1 achieves state-of-the-art performance on long video benchmarks, including VideoMME, LongVideoBench, and MLVU. Notably, it significantly improves the genuine rectification rate, demonstrating the high reliability of our proposed reflection paradigm.

2 Can MLLM Correct Itself? A Preliminary Investigation

To investigate the self-correction capability of existing multimodal large language models in long video understanding, we conduct a preliminary empirical study on the LongVideoBench and VideoMME datasets. As shown in Fig. 2, we observe a counterintuitive phenomenon where relying solely on internal parametric knowledge for multi-turn reflection, transitioning from the initial intuition to the final response, fails to yield performance improvements and instead leads to a significant drop in accuracy. Specifically, the base model Qwen2.5-VL-7B experiences a sharp decline in accuracy on both benchmarks, dropping from 48.59% to 41.96% on LongVideoBench. Furthermore, even recent open-source models optimized

specifically for multimodal reasoning, such as Video-R1-7B and Vision-R1, generally suffer from performance degradation during this purely internal reflection process.

To diagnose the root cause of this performance degradation, we analyze the behavioral logic of the models during the reflection process. In long video scenarios featuring extremely high visual information density and large temporal spans, the initial intuition of the model is highly susceptible to factual deviations due to missing keyframes or truncated contexts. Under these circumstances, forcing the model to self-correct without acquiring new visual evidence often causes it to over-rely on its initially generated erroneous context. This closed-loop reflection lacking external visual anchors prevents the model from establishing rigorous objective verification standards. Instead of accurately locating and correcting errors, the model tends to perpetuate or even amplify the initial visual hallucinations during repeated internal reasoning loops, ultimately leading to an overall accuracy decline where $\Delta \text{Acc} < 0$.

The above analysis exposes the core limitation of closed-loop reflection where models lack independent and objective external visual evidence as a verification standard during the reasoning process. This insight directly motivates the core design of Reflect-R1. Relying strictly on internal knowledge prevents models from catching their own errors. To fix this, we must introduce external tools so the model can actively retrieve fresh evidence. Building upon this argument, we propose Reflect-R1 to achieve genuine self-correction by empowering the model with the capability to autonomously collect objective evidence and rigorously verify facts.

3 Methodology

3.1 Problem Formulation and Inference Framework

Given a long video V and a natural language question q , the objective is to generate an accurate textual response y . We model this procedure as a multi-step reasoning chain. Diverging from conventional approaches that directly approximate the single mapping distribution $P(y|V, q)$, we propose a dynamic decision-making process incorporating intuition, independent verification, and arbitration.

The core philosophy of Reflect-R1 is to train a single unified policy π_θ that internalizes intuition, verification, and arbitration behaviors, rather than training multiple independent sub-models. Operating within a multi-stage Markov Decision Process, the policy exhibits distinct reasoning behaviors conditioned on the current context state.

In the intuition stage, conditioned on the raw video V and question q , the policy leverages parametric intuition to rapidly generate an initial answer y_1 while autonomously invoking retrieval tools to localize a set of keyframes F .

$$y_1, F \sim \pi_\theta(\cdot|V, q). \quad (1)$$

During the verification stage, we enforce strict contextual isolation to ensure an independent evaluation. In this phase, the policy is denied access to the initial

hypothesis y_1 and the global video V , with its input scope strictly restricted to a local context comprising only the question q and the retrieved keyframes F . Relying exclusively on this retrieved evidence, the model generates an independent verification response y_2 , thereby providing an unbiased assessment.

$$y_2 \sim \pi_\theta(\cdot|F, q). \quad (2)$$

In the arbitration stage, acting as the final arbitrator, the policy synthesizes the intuitive hypothesis y_1 and the independent verification y_2 . To guarantee robustness, π_θ employs an active investigation mechanism where the model, regardless of consensus between y_1 and y_2 , is mandated to re-invoke tools to backtrack through video V for deep evidentiary re-confirmation, ultimately yielding the final answer y_3 .

$$y_3 \sim \pi_\theta(\cdot|V, F, q, y_1, y_2). \quad (3)$$

While these behaviors are executed by a shared parameter set θ , the task difficulty and reward scales vary significantly across stages. To address this, we propose the Stage-Decoupled GRPO algorithm, which fully decouples the advantage estimation for each stage during training. As detailed in Sec. 3.3, this design prevents cross-stage competition during optimization.

3.2 The Dilemma of End-to-End Optimization

Before detailing the stage-decoupled GRPO (SD-GRPO) algorithm, we analyze why a naive end-to-end optimization paradigm fails to elicit genuine self-correction capabilities in long video question answering. Specifically, the end-to-end approach attempts to simultaneously train intuition (y_1), verification (y_2), and arbitration (y_3) behaviors within a single training phase. As Figure 3 illustrates, we compare the training dynamics of this joint end-to-end strategy against our decoupled method.

Under the joint end-to-end training regime (Fig. 3 (a)), the model inevitably suffers from policy coupling. Because all reasoning stages undergo drastic gradient updates simultaneously, the arbitration policy π_θ tends to exploit optimization shortcuts. To rapidly secure base rewards, the policy directly copies the initial intuitive hypothesis y_1 instead of learning the complex logic required for error correction. This phenomenon eliminates the performance gap between y_1 and y_3 , strips the model of its error-correction utility, and causes the arbitration behavior to degenerate into a trivial identity mapping.

In contrast, our decoupled training strategy (Fig. 3 (b)) effectively resolves this issue through a two-stage design. By first stabilizing the intuition phase, we ensure that the subsequent arbitration stage learns to correct a stable set of initial errors. Empirical results demonstrate that the policy π_θ develops genuine error-correction capabilities only when the initial reasoning process remains stable, rather than simply memorizing the final answers. This finding directly confirms the necessity of the decoupled architecture in SD-GRPO, which relies on a stable foundation to unlock authentic self-reflection.

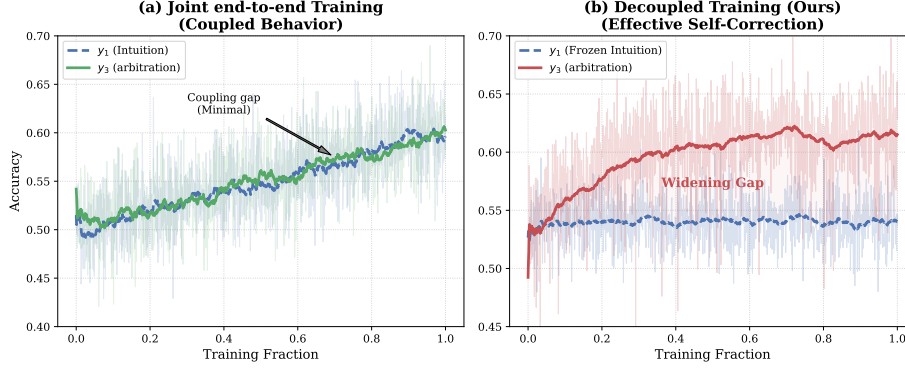


Fig. 3: Decoupled training prevents policy coupling. (a) Joint end-to-end training collapses reflection into a trivial identity mapping of the initial intuition. (b) Our decoupled strategy stabilizes the preceding distribution, enabling the model to learn robust error-correction logic and achieve a widening performance gap.

3.3 Stage-Decoupled GRPO (SD-GRPO)

Building upon the inference framework established in Sec. 3.1, we formalize the multi-step reasoning process as the sequential generation of three variables: intuition y_1 , independent verification y_2 , and arbitration y_3 . To effectively optimize this long-chain reasoning process through GRPO [27], we propose the Stage-Decoupled GRPO (SD-GRPO) algorithm, as illustrated in Fig. 4. This method addresses the credit assignment [1, 18, 21] challenge inherent in varying reasoning depths through a group-wise advantage decoupling mechanism, while integrating stage-aware rewards and a progressive two-stage optimization to facilitate robust evolution from intuition to arbitration.

Unified Objective and Advantage Decoupling. The overall optimization objective aims to maximize the cumulative expected return across the three reasoning stages. Formally, we define the total loss function $\mathcal{L}(\theta)$ as a weighted summation of the GRPO objectives corresponding to y_1 , y_2 , and y_3 :

$$\mathcal{L}(\theta) = \sum_{k=1}^3 \mathbb{E}_{q \sim \mathcal{D}} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{L_{i,k}} \sum_{t=1}^{L_{i,k}} \left(\mathcal{J}_{\text{clip}}^{(k)}(t, i) - \beta_k \mathbb{D}_{\text{KL}} \left(\pi_{\theta}(\cdot | x^{(k)}) \| \pi_{\text{ref}}(\cdot | x^{(k)}) \right)_t \right) \right], \quad (4)$$

where $\mathcal{J}_{\text{clip}}^{(k)}$ represents the PPO-based [26] clipped surrogate objective designed to stabilize policy updates:

$$\mathcal{J}_{\text{clip}}^{(k)}(t, i) = \min \left(\rho_{t,i}^{(k)} A_i^{(k)}, \text{clip}(\rho_{t,i}^{(k)}, 1 - \epsilon, 1 + \epsilon) A_i^{(k)} \right). \quad (5)$$

Here, $\rho_{t,i}^{(k)} = \frac{\pi_{\theta}(y_{t,i} | x^{(k)})}{\pi_{\text{old}}(y_{t,i} | x^{(k)})}$ denotes the importance sampling ratio, and $A_i^{(k)}$ is the advantage term computed via group-wise normalization. The index $k \in \{1, 2, 3\}$

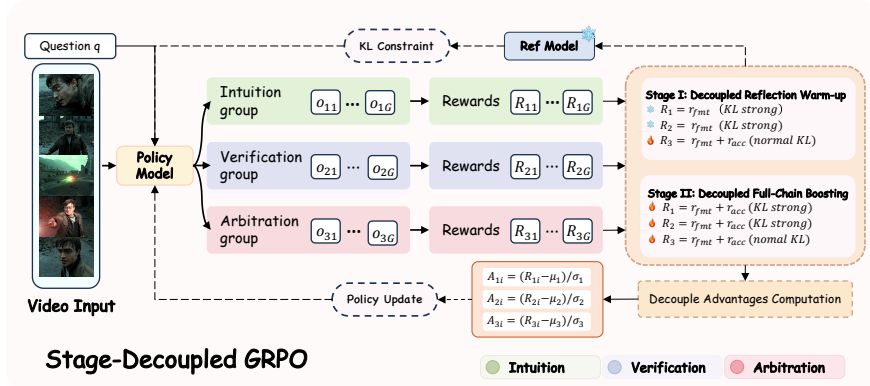


Fig. 4: Overview of SD-GRPO. We employ a progressive two-stage optimization: Stage I warms up the arbitration policy via strong KL constraints, and Stage II performs full-chain joint optimization. Additionally, group-wise advantage decoupling ensures that each reasoning stage evolves independently.

corresponds to the generation processes for y_1, y_2 , and y_3 respectively. Accordingly, $x^{(k)}$ represents the stage-specific input context established in Sec. 3.1, where $x^{(1)} = \{V, q\}$, $x^{(2)} = \{F, q\}$, and $x^{(3)} = \{V, F, q, y_1, y_2\}$. The term G represents the group size used for sampling, and β_k controls the strength of the KL divergence penalty to prevent excessive deviation from the reference policy π_{ref} .

A critical challenge in multi-step reasoning is the significant disparity in task difficulty between the intuition (y_1) and arbitration (y_3) stages. Global normalization often causes the simpler intuition stage to dominate gradient updates because it inherently yields higher baseline rewards. To resolve this issue, we compute advantages independently within each generation stage. Specifically, for the i -th sample in the k -th stage, the advantage $A_i^{(k)}$ is defined as follows:

$$A_i^{(k)} = \frac{R_i^{(k)} - \mu_k}{\sigma_k + \epsilon}, \quad (6)$$

where μ_k and σ_k are derived exclusively from the reward set $\{R_1^{(k)}, \dots, R_G^{(k)}\}$ associated with that specific stage. This design establishes a principle of intra-stage competition, where y_1 samples are compared solely against other y_1 samples, and similarly for y_3 . This mechanism effectively isolates the reward distributions across different reasoning depths, ensuring that subtle improvement signals in the arbitration phase are not overshadowed by the high baselines inherent to the intuition phase.

Stage-Aware Reward Function. To elicit differentiated reasoning behaviors across y_1, y_2 , and y_3 , we design a fine-grained reward function $R^{(k)} = r_{fmt} + r_{acc}^{(k)}$, where r_{fmt} represents a universal format constraint reward and $r_{acc}^{(k)}$ is tailored to the specific characteristics of each stage.

The format reward r_{fmt} serves as a structural regularizer to ensure syntactic correctness across all outputs. Specifically, it aggregates constraints from three dimensions: 1) Tag Adherence, which enforces the proper usage of XML

delimiters; 2) Thought Length Reward, which encourages sufficient deliberation by regulating the length of the reasoning chain; and 3) Valid Tool Invocation, which verifies the syntactic accuracy and executability of API calls. Due to space constraints, the detailed mathematical formulations and implementation details of these format rewards are provided in the Appendix.

For the intuition stage ($k = 1$), we employ a standard binary reward where $r_{\text{acc}}^{(1)} = \mathbb{I}(y_1 = y_{gt})$. In the verification stage ($k = 2$), we introduce an honesty incentive to ensure the objectivity of the evaluation. Because y_2 is generated under a severely restricted field of view containing only the retrieved keyframes F , applying a standard binary reward would inevitably force the model into blind guessing when the provided visual evidence is insufficient. As recent studies on model abstention demonstrate [8, 20, 32, 38], forcing responses under partial information significantly exacerbates hallucinations. To address this, we design a ternary reward mechanism:

$$r_{\text{acc}}^{(2)} = \begin{cases} 1, & \text{if } y_2 = y_{gt} \\ 0, & \text{if } y_2 \in \mathcal{S}_{\text{abstain}} \\ -1, & \text{otherwise} \end{cases}, \quad (7)$$

where $\mathcal{S}_{\text{abstain}}$ denotes the set of abstention responses (e.g., "I don't know"). This mechanism explicitly incentivizes the model to acknowledge ignorance when visual clues are inadequate, which yields a neutral score. It effectively penalizes errors resulting from baseless fabrication, thereby guaranteeing that the verification process remains strictly grounded in observable empirical evidence.

For the arbitration stage ($k = 3$), we implement an anti-corruption penalty to prevent the model from overturning originally correct judgments. The reward is formulated as follows:

$$r_{\text{acc}}^{(3)} = \begin{cases} 1, & \text{if } y_3 = y_{gt} \\ -1, & \text{if } y_3 \neq y_{gt} \wedge (y_1 = y_{gt} \vee y_2 = y_{gt}) \\ 0, & \text{otherwise} \end{cases}. \quad (8)$$

This structure ensures that if the final answer y_3 is incorrect while at least one of the preceding outputs y_1 or y_2 is correct, the model incurs a strict penalty. This constraint compels the policy to exercise extreme caution and avoid destructive modifications during the final arbitration.

3.4 Training Strategies

Data construction. We aggregate video and question-answer pairs from six datasets: LLaVA-Video-178K [46], Panda-70M [3], NExT-QA [41], Perception-Test [24], CLEVRER [44], and STAR [39]. We employ Qwen2.5-VL-72B [2] to synthesize chain-of-thought data aligned with our three-stage pipeline. A rule-based filter eliminates defective outputs to generate the Reflect-R1-CoT-90k dataset for SFT. Finally, GPT-4o [13] selects 30,000 challenging samples to construct the Reflect-R1-RL-30k dataset for reinforcement learning. The appendix provides additional details.

Table 1: Long video understanding performance. We compare our method against state-of-the-art models. The baselines are categorized into two primary paradigms: standard inference relying solely on internal parameters (w/o Tools), and tool-augmented reasoning. [†] indicates keyframes adaptively retrieved during the inference process.

Model	# Frame	VideoMME (w/o sub)				MLVU	LVB
		short	medium	long	overall	m-avg	val
<i>w/o Tools</i>							
Qwen2.5-VL-7B-Instruct [2]	768	71.4	60.1	52.3	61.3	57.9	53.9
GPT-4o [13]	384	80.0	70.3	65.3	71.9	64.6	66.7
Gemini-1.5-Pro [31]	1 fps	81.7	74.3	67.4	75.0	-	64.0
VL-Rethinker [33]	768	70.1	60.8	53	61.3	63.5	55.58
Vision-R1-7B [12]	768	71.2	60.2	52.8	61.4	68.2	52.8
Video-R1-7B [6]	768	72.2	58.1	52.3	60.8	68.5	60.1
<i>Tool-Augmented Reasoning</i>							
VideoAgent (GPT-4) [35]	87 [†]	-	-	49.0	56.0	-	-
T* (GPT-4o) [43]	32 [†]	72.1	60.3	52.0	61.45	-	-
TimeSearch-R-7B [22]	768	73.5	61.2	53.4	62.7	69.3	61.2
Reflect-R1 (Ours)	768	73.9	61.0	55.6	63.5	69.8	62.5

Model training. Reflect-R1 employs a two-stage training pipeline. First, supervised fine-tuning provides a cold start for the model to acquire structured reasoning formats and fundamental reflection paradigms. Subsequently, our SD-GRPO algorithm performs reinforcement learning post-training to unlock deep reasoning capabilities, enabling autonomous tool invocation and dynamic self-correction.

4 Experiments

4.1 Setup

Benchmarks. We evaluate the proposed method on three widely adopted long-form video benchmarks: VideoMME [7], LongVideoBench [40] and MLVU [48].

Training Details. We train the model using 8 NVIDIA H200 GPUs. During the training phase, we limit the maximum number of video frames to 734, processing each frame at a maximum resolution of $192 \times 28 \times 28$ pixels. During inference, we increase the frame capacity to 768 while maintaining all other configurations. For the reinforcement learning process, we set the group size G to 8. More details are provided in Appendix.

4.2 Main Results

Long-Form Video Understanding. Reflect-R1 demonstrates remarkable performance across multiple challenging long-form video understanding benchmarks,

Table 2: Temporal search performance. We report temporal similarity, visual similarity, and question-answering (QA) accuracy on Haystack-LVBench. Baseline results are directly cited from [43]. [†] indicates the average number of keyframes determined by the model adaptively.

Method	Base Model	# Frame	Temporal			Visual			QA
			P↑	R↑	F ₁ ↑	P↑	R↑	F ₁ ↑	LVBench
Static Frame Sampling									
Uniform	Qwen2.5VL-7B	8	1.4	6.3	2.2	56.0	72.0	62.7	33.7
Uniform	GPT-4o	8	1.4	6.3	2.2	56.0	72.0	62.7	47.1
Uniform	GPT-4o	32	1.4	24.9	2.7	58.7	81.6	67.3	50.5
Adaptive Temporal Search									
VideoAgent [35]	GPT-4	10.1 [†]	1.2	8.5	2.1	58.8	73.2	64.7	-
Retrieval-based [43]	GPT-4o	8	1.5	6.3	2.3	63.1	65.5	64.1	-
T* [43]	GPT-4o	8	1.6	7.1	2.5	58.4	72.7	64.3	51.9
Retrieval-based [43]	GPT-4o	32	1.3	21.8	2.4	59.9	80.8	67.8	-
T* [43]	GPT-4o	32	1.7	28.2	3.1	58.3	83.2	67.8	53.1
Active Tool-Augmented Search									
TimeSearch-R [22]	Qwen2.5VL-7B	9.32 [†]	5.5	21.2	8.1	63.2	76.5	69.2	51.5
Reflect-R1 (Ours)	Qwen2.5VL-7B	10.18 [†]	6.3	21.9	8.9	63.8	76.1	69.5	55.5

as shown in Table 1. Reflect-R1 outperforms baseline models such as Qwen2.5-VL-7B by capturing definitive visual evidence through dynamic temporal search and performing reflective self-correction. Experimental results indicate that the performance margin of Reflect-R1 expands as the video duration increases, validating the robustness of its decoupled verification paradigm in processing complex long-temporal contexts.

Temporal Search. As shown in Table 2, Reflect-R1 outperforms state-of-the-art baselines in temporal similarity, visual similarity, and question-answering accuracy. This performance leap stems from our Evidence-Driven dynamic invocation mechanism. Unlike traditional single-pass retrieval pipelines, the arbitration stage proactively identifies verification failures caused by insufficient evidence and triggers tool re-invocation to retrieve the correct frames. Through the stage-decoupled optimization of SD-GRPO, the model learns an iterative and goal-oriented search strategy. This closed-loop feedback ensures that the retrieved visual evidence effectively supports complex reasoning, driving the synergistic evolution of keyframe localization precision and genuine self-correction capabilities.

Reflection Reliability. As shown in Table 3, we evaluate self-correction efficacy by comparing initial and final accuracy. Baselines relying solely on internal parameters generally suffer from performance degradation ($\Delta \text{Acc} < 0$). Without external visual anchors, closed-loop reflection amplifies visual hallucinations and frequently overturns initially correct judgments. In contrast, Reflect-R1 employs dynamic tool-use reflection to break this hallucination loop, achieving consistent accuracy improvements across benchmarks. This demonstrates that independent

Table 3: Reflection Reliability Analysis. We evaluate the self-correction capabilities across paradigms by comparing the initial intuitive accuracy and the final accuracy.

Model	LongVideoBench			VideoMME		
	Init	Final	Δ Acc	Init	Final	Δ Acc
<i>Internal Reflection (w/o Tools)</i>						
Qwen2.5-VL-7B (Base) [2]	48.59%	41.96%	-6.63%	60.96%	54.00%	-6.96%
VL-Rethinker [33]	53.87%	46.63%	-7.24%	61.32%	53.81%	-7.51%
Vision-R1 [12]	52.88%	52.15%	-0.73%	61.37%	61.45%	+0.08%
Video-R1-7B [6]	59.88%	58.53%	-1.35%	60.44%	59.59%	-0.85%
<i>Evidence-Driven Reflection</i>						
Reflect-R1 (Ours)	59.68%	62.50%	+2.82%	60.28%	61.69%	+1.41%

Table 4: Training strategy ablation on LongVideoBench. **Table 5:** Reward function ablation on LongVideoBench.

Strategy Variant	Init (y_1)	Verif (y_2)	Arbitration (y_3)
Joint end-to-end GRPO	61.2	57.3	61.8
w/o Verification (y_2)	59.1	-	55.9
w/o Advantage Decoupling	62.1	57.4	61.4
Reflect-R1 (Full)	59.68	56.62	62.50

Reward Variant	Init (y_1)	Verif (y_2)	Arbitration (y_3)
w/o Abstention Incentive (y_2)	59.2	61.2	60.8
w/o Anti-Corruption Penalty (y_3)	61.8	59.2	61.3
Reflect-R1 (Full Rewards)	59.68	56.62	62.50

visual verification is crucial for overcoming reflection degradation in large vision-language models.

4.3 Ablation Study

Component Ablation. Tables 4 and 5 detail our ablation study on the LongVideoBench.

Regarding training strategies, joint end-to-end optimization yields minimal performance gains from the intuition phase to final arbitration (from 61.2% to 61.8%), confirming that coupled training induces policy collapse. Bypassing the independent verification response y_2 while retaining active tool invocation fails to break the hallucination loop, causing the final accuracy to plummet to 55.9%. Furthermore, replacing group-wise advantage decoupling with global calculation causes the reflection process to actually degrade arbitration accuracy below the model’s own initial intuition (from 62.1% down to 61.4%), as simple intuitive tasks overshadow the complex optimization signals required for reflection. For reward formulation, ablating the abstention incentive forces the model to guess blindly under information bottlenecks, which prevents objective answer verification and misleads the final reflection, stalling the final accuracy at 60.8%. Similarly, removing the anti-corruption penalty causes the model to frequently overturn initially correct judgments, resulting in an accuracy of 61.3%. The complete Reflect-R1 framework integrates these stage-decoupled optimizations and fine-grained rewards to achieve the highest final accuracy of 62.50%.

Analysis of Training Dynamics. Fig. 5 illustrates the training trajectory over the first 500 steps. The distinct stratification of the reward curves (b) demonstrates that SD-GRPO successfully decouples the reasoning stages and effectively prevents policy coupling. The accuracy of the blind-test verification (y_2) experiences an initial decline, yet its corresponding reward climbs steadily

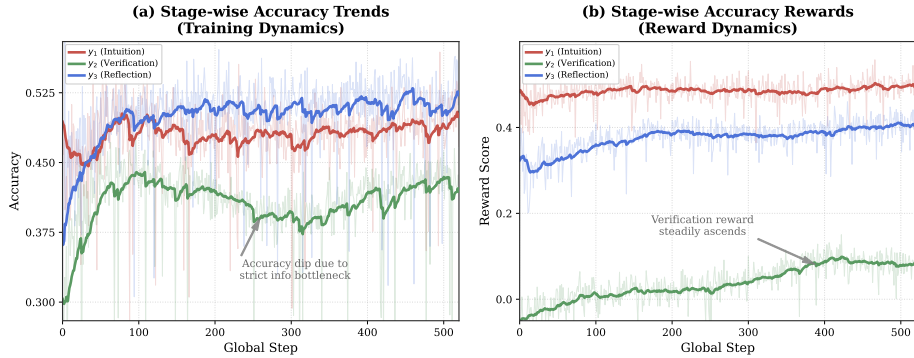


Fig. 5: Training Dynamics of Stage-Decoupled Verification.

from a negative value (a). This phenomenon validates the effectiveness of our reward design. Constrained by the information bottleneck, the model learns to abandon speculative guessing and instead formulates answers strictly based on visual evidence. Furthermore, despite the decoupled policies, the demand of y_2 for high-quality evidence acts as environmental feedback. This feedback compels the intuitive stage (y_1) to continuously improve its retrieval precision, which drives a synergistic evolution between the two stages. Ultimately, the reflective arbitration (y_3) utilizes this reliable evidence to correct initial errors and establishes a widening performance gap with y_1 . This dynamic conclusively proves the emergence of genuine self-correction capabilities within the model.

5 Qualitative Analysis

Figure 6 illustrates the autonomous self-correction of Reflect-R1. When the initial intuition (y_1) suffers from visual hallucinations, the independent verification (y_2) provides objective counter-evidence. The reflective arbitration (y_3) then resolves this conflict through adaptive temporal search to derive the correct answer. This process confirms that SD-GRPO effectively breaks hallucination loops via evidence-based reasoning.

6 Limitations

Although Reflect-R1 performs well on long-video reasoning, it has several limitations. First, the three-stage procedure often requires multiple tool calls and long generations, which increases inference latency and compute cost, so it is less suitable for strict real-time settings. Second, the method relies on temporal retrieval to obtain evidence. When retrieval is inaccurate or the key cues are weak, the subsequent verification and arbitration stages may have insufficient support to correct errors. Finally, overall performance still depends on the underlying vision-language model, which can fail on a small number of cases with

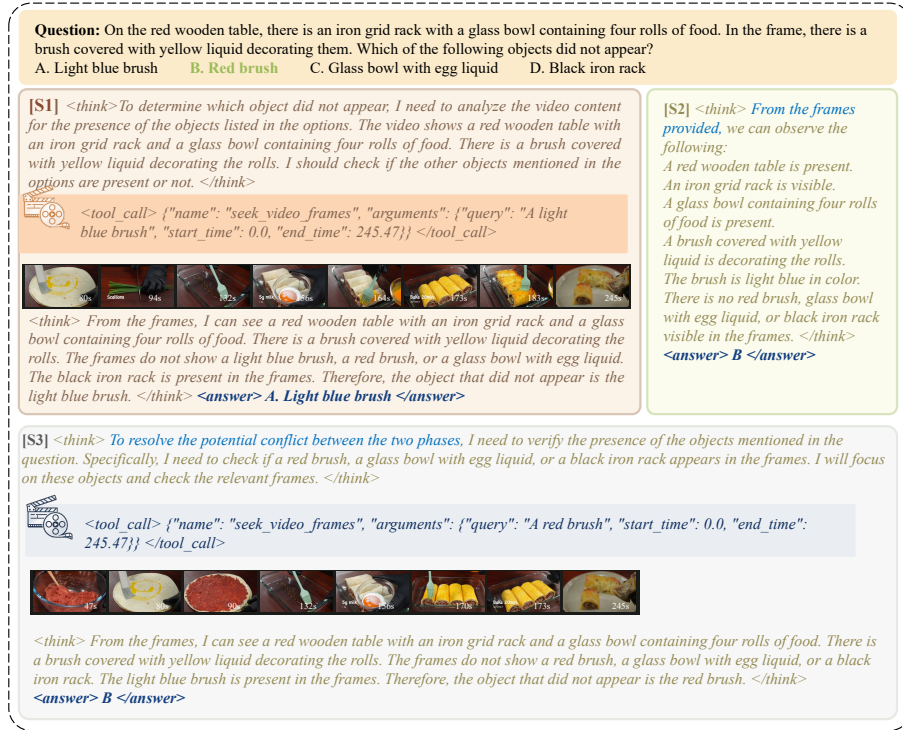


Fig. 6: Qualitative example of the reflective reasoning process.

high complexity or ambiguous evidence. Future work focuses on improving inference efficiency and integrating a broader set of more accurate tools to strengthen evidence acquisition and robustness.

7 Conclusion

In this work, we propose Reflect-R1, an Evidence-Driven reflection framework integrating stage-decoupled verification and dynamic tool invocation to address the self-correction challenge in long video understanding. To overcome the Internal Closed-Loop Reflection and the policy coupling in multi-stage reinforcement learning, we design the SD-GRPO algorithm. This algorithm drives the synergistic evolution of intuition, verification, and arbitration through intra-group advantage isolation. Reflect-R1 achieves state-of-the-art performance on benchmarks including VideoMME, LongVideoBench and MLVU, enabling genuine self-correction strictly grounded in objective evidence. We hope this work makes a meaningful contribution to advancing reinforcement learning for reflection in multimodal large language models.

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Reflect-R1: Evidence-Driven Reflection for Self-Correction in Long Video Understanding

Supplementary Material

This appendix provides additional details of our method. It is organized as follows:

- Section A: Related Work
- Section B: Reward Details
- Section C: Dataset Construction Details
- Section D: Training Details
- Section E: Prompt Design
- Section F: More Case Studies

A Related Work

Reflective reasoning has emerged as a prominent research direction to enhance the reliability of large language models. Early efforts predominantly focus on self-feedback and iterative refinement within the pure text domain. For example, Self-Refine [19] generates feedback from the same model to iteratively rewrite the output, whereas Reflexion [29] utilizes reflective memory in natural language to improve subsequent sequential decisions. These approaches demonstrate that models can significantly improve output quality through self-examination during the reasoning phase. However, their core mechanisms remain constrained by closed-loop modifications based on internal representations, lacking explicit reliance on external objective facts. ReAct [42] further intertwines reasoning and acting, enabling models to invoke external tools or knowledge sources during the thought process. This establishes the foundation for subsequent agentic reasoning and tool-augmented mechanisms.

In the field of multimodal reasoning, recent studies begin to integrate reflection mechanisms and reinforcement learning into vision-language models. Works such as Vision-R1 [12] and Video-R1 [6] explore how to directly elicit the long-chain reasoning and alignment capabilities of large multimodal models through vision-guided reinforcement learning. Furthermore, VL-Rethinker [33] explicitly encourages self-reflection in models using reinforcement learning. These studies demonstrate the substantial potential of reflection mechanisms and reinforcement learning in multimodal scenarios. Nonetheless, they primarily target static images or general visual question answering. They fail to systematically address the coordination challenges of evidence retrieval, independent verification, and final arbitration caused by massive information streams in long video contexts. In contrast, we focus on evidence-driven self-correction in long video understanding. We extend the reflection paradigm from a closed-loop review completely dependent on internal parameters to a rigorous verification process grounded in external temporal objective evidence.

Another research trajectory closely related to our work involves tool-augmented reasoning, structured representation, and active temporal retrieval in long video

understanding. VideoAgent [35] treats the large model as a central agent to collect key information in videos through iterative planning and tool invocation. From a representation perspective, VideoTree [36] proposes a query-adaptive hierarchical tree structure to compress redundant information in a coarse-to-fine manner, thereby supporting efficient long-temporal reasoning. Additionally, the representative work by Ye et al. [43] points out that the core difficulty of long video understanding lies in accurately locating a minimal amount of key evidence from massive frames, and accordingly proposes the T* temporal retrieval framework. Concurrently, TimeSearch-R [22] combines adaptive temporal retrieval with a self-verification mechanism, emphasizing the decisive role of retrieval quality in subsequent reasoning. These methods collectively indicate that relying solely on static inputs or one-time sampling usually encounters performance bottlenecks when processing long videos. Models must possess the capability to actively search, compress, and organize relevant segments. Our work aligns with them in emphasizing the importance of external retrieval. Advancing this concept, we propose a three-stage framework comprising intuition, verification, and arbitration. By explicitly decoupling the optimization objectives of different cognitive stages through SD-GRPO, we unify tool invocation, objective evidence verification, and final noise-resistant arbitration within a single evidence-driven reflection framework.

B Reward Details

B.1 Format Reward Details

As introduced in Section 3.3, we incorporate a format reward (r_{fmt}) to constrain the behavior of the policy model during multi-stage reasoning. This reward ensures structural consistency in the outputs and prevents reward hacking, a prevalent issue in reinforcement learning. The format reward linearly combines three fine-grained constraint components: tag adherence, thought length reward, and valid tool invocation reward.

Tag Adherence. We impose strict XML format constraints on the model outputs. The responses must include explicit reasoning tags, specifically `<think>` and `</think>`. These must be followed by either answer tags (`<answer>` and `</answer>`) or tool invocation tags (`<tool_call>` and `</tool_call>`). We formulate a rigorous binary format detection function r_{tag} using indicator variables for correct tag closures. Let $\mathcal{I}_{think}$, $\mathcal{I}_{ans}$, and $\mathcal{I}_{tool}$ represent the successful parsing of their respective tag pairs. The reward is defined as:

$$r_{tag} = \begin{cases} 1.0, & \text{if } \mathcal{I}_{think} = 1 \text{ and } (\mathcal{I}_{ans} \oplus \mathcal{I}_{tool}) = 1, \\ 0.0, & \text{otherwise.} \end{cases}$$

where $\oplus$ denotes the logical exclusive OR operation. If the model mixes tool invocations and final answers in the same turn or exhibits tag closure errors, the sequence is immediately truncated and receives the zero reward.

Thought Length Reward. In long-chain reasoning, excessively short thoughts often indicate that the model is taking optimization shortcuts to guess blindly. Conversely, unconstrained and verbose reasoning inevitably introduces hallucinations and wastes computational resources. To address this, we design a piecewise length reward mechanism featuring soft boundaries and an asymmetric penalty. Let l represent the length of the text generated within the `<think>` tags, while L_{min} and L_{max} denote the lower and upper bounds of the optimal reasoning length. In our experiments, these bounds are set to 120 and 700 respectively. The length reward r_{len} is formulated as follows:

$$r_{len} = \begin{cases} l/L_{min}, & \text{if } 0 < l < L_{min}, \\ 1.0, & \text{if } L_{min} \leq l \leq L_{max}, \\ 1.0 - \frac{l - L_{max}}{L_{max}}, & \text{if } L_{max} < l \leq 2L_{max}, \\ -1.0, & \text{if } l > 2L_{max}. \end{cases}$$

Valid Tool Invocation Reward. We design a tool invocation incentive specifically for the intuition and arbitration stages, as both phases require the active acquisition of objective evidence. To prevent reward hacking where the model exploits system fallback parameters to accumulate tool-use scores, we introduce posterior verification based on execution results. Let a_{tool} denote the parsed arguments of the tool call and $E(a_{tool})$ represent the execution feedback indicating the number of successfully retrieved frames. The valid tool invocation reward r_{tool} follows a strict binary formulation:

$$r_{tool} = \begin{cases} 1.0, & \text{if parsing succeeds and } E(a_{tool}) > 0, \\ 0.0, & \text{otherwise.} \end{cases}$$

This ensures that a tool invocation is deemed valid only when it returns actual visual information.

B.2 The Necessity of the Abstention Incentive in Verification

Section 3.3 introduces a ternary reward mechanism with an abstention incentive for the independent verification stage. We illustrate the necessity of this design through a diagnostic failure case in Fig. 7. Under a standard binary reward formulation, forcing the model to guess severely impairs the training process. If a baseless guess happens to match the ground truth, the model receives a reward. This feedback reinforces lucky guessing behaviors and prevents the model from conducting objective verification based on the available visual evidence. Consequently, the model generates misleading answers that interfere with the third-stage arbitration. The ternary abstention reward eliminates this issue by allowing the model to abstain and receive a neutral reward when evidence is insufficient. This mechanism forces the verification stage to act as a rigorous evidence reviewer. More importantly, an honest abstention explicitly triggers the

arbitration stage to reinvoke temporal search tools. By actively mining new visual clues from the global video, the framework successfully completes a genuine closed-loop verification.

[S2] <think>Based on the provided video frame at 45.2s, I can observe a man standing in a kitchen holding a closed blue box. The question asks 'What ingredient does the man take out of the blue box?'. Looking closely at the visual evidence, the box is completely opaque and closed in this specific frame. The man's hands are simply resting on the lid. **There is no visual information** showing the box being opened or any ingredient being removed. Therefore, **the provided frame lacks sufficient evidence to determine the exact ingredient.**</think><answer>B</answer>

Fig. 7: A disconnect between reasoning and response occurs during the blind verification stage (S2) due to the absence of an abstention incentive. The model accurately identifies the lack of visual evidence during its reasoning process but is forced to make an ungrounded guess in the final output.

C Dataset Construction Details

Fig. 8 illustrates the characteristics of the training data. The data sources primarily consist of LLaVA-Video (48.5%) and Panda-70M (27.3%), supplemented by STAR, NeXT-QA, CLEVRER, and PerceptionTest. Additionally, the video duration distribution spans a broad range with a mean of 641 seconds, which provides a solid foundation for evaluating long-context reasoning capabilities. Qwen2.5-VL-72B is used to generate the supervised fine-tuning data. For each question, the model concurrently produces responses for the intuition, verification, and arbitration stages. An initial filtering step is then applied based on the correctness of these generated responses. The intuition and arbitration subsets are further refined by retaining only samples that contain valid tool invocations. A balanced proportion across the three stages is maintained throughout the construction process. For reinforcement learning data selection, GPT-4o receives four frames sampled from a long video to answer the corresponding question. Questions that can be answered correctly under this limited visual context are regarded as invalid and the associated long-video samples are discarded.

D Training Details

We employ a cosine learning rate decay schedule for both the supervised fine-tuning cold start and the reinforcement learning phases. The model undergoes supervised fine-tuning for a single epoch. Subsequently, we conduct the first stage of reinforcement learning to stabilize the reflection capabilities and fundamental output formatting. This is followed by the second stage of reinforcement learning. During the optimization process, the Kullback-Leibler divergence coefficients β_1 , β_2 , and β_3 for the three reasoning stages are set to 0.05, 0.05, and 0.005, respectively.

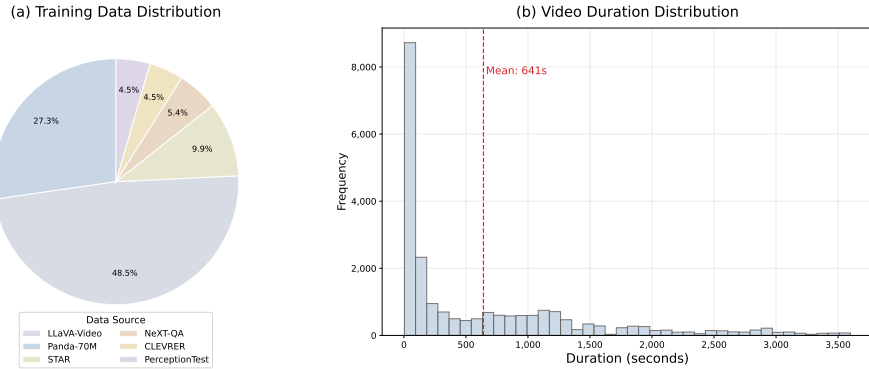


Fig. 8: Training data distribution across different source datasets and video durations.

E Prompt Design

This section details the specific prompts utilized across the system level and the three distinct reasoning stages. These prompts rigorously regulate the output formatting, tool invocation protocols, and the abstention mechanism of the model. This section also includes the reflection prompt used by the internal closed-loop reflection baseline for comparison.

System Prompt. We follow the official tool-use protocol of the Qwen2.5-VL series and adopt its `<tool_call>` format to invoke the temporal retrieval tool, as shown in Fig. 9. This design standardizes the instruction format for tool invocation, allowing the model to generate executable retrieval commands and thereby obtain the video frames required for arbitration with precision.

Intuition Prompt. During the first stage, the prompt mandates that the model must invoke the retrieval tool at least once to explore the video content before providing a final answer, as detailed in Fig. 10. The policy is permitted a maximum of 10 interaction rounds, and it is strictly required to conduct chain-of-thought reasoning within designated tags prior to executing any action.

Verification Prompt. The second-stage prompt severely restricts the context, compelling the model to base its reasoning exclusively on the provided local image frames (see Fig. 11). Crucially, this prompt integrates the abstention mechanism by explicitly instructing the model to output an acknowledgment of ignorance when visual information is insufficient. This instruction effectively prevents forced hallucination under information bottlenecks.

Arbitration Prompt. The prompt for the third stage presents the model with the complete interaction history of the preceding phases, as shown in Fig. 12. It forces the policy to explicitly extract and quote the conclusions from the first two stages at the beginning of its reasoning process. Furthermore, it strictly prohibits direct answering in the initial turn, compelling the model to autonomously invoke tools to verify key evidence independently, which facilitates genuine self-correction.

Reflection Prompt (Baseline). For comparison, we also construct an internal closed-loop reflection baseline that revisits the model’s previous answer using

System Prompt

You are a critical reasoning arbitrator capable of analyzing multiple reasoning paths and deriving the most accurate conclusion through tool-assisted verification.

Tools

You may call one or more functions to assist with the user query.

You are provided with function signatures within <tools></tools> XML tags:

```
<tools>
{"type": "function", "function": {"name": "seek_video_frames", "description": "Search and select video frames according to textual query and temporal window. Time is in seconds.", "parameters": {"type": "object", "properties": {"query": {"type": "string", "description": "The query is used to describe the object, scene, or event of interest in the video thoroughly and clearly."}, "start_time": {"type": "number", "description": "Start time of the segment of interest."}, "end_time": {"type": "number", "description": "End time of the segment of interest."}, "num_frames": {"type": "integer", "description": "Number of frames to sample (maximum 8). Default is 8."}}, "required": ["query"]}}}
</tools>
```

For each function call, return a json object with function name and arguments within <tool_call></tool_call> XML tags:

```
<tool_call>
{"name": <function-name>, "arguments": <args-json-object>}
</tool_call>
```

Fig. 9: The system prompt used to define the model persona and tool signatures.

Intuition Prompt

You must conduct reasoning inside <think> and </think> first every time you call the tool or answer the question. You must invoke tools to explore any video content you are interested in within <tool_call> </tool_call> tags.

You cannot answer the question directly. You are required to invoke the tool at least once to explore the video content before providing your final answer.

You will then receive the tool response along with the corresponding video frames. You must wait for the response from the tool before answering or invoking the tool again.

If you are not sure, invoke again before answering. You are allowed to use <tool_call></tool_call> tags for a maximum of 10 rounds.

When you have enough information to answer the question, provide your answer within <answer> </answer> tags. Your answer should be supported by evidence from the video.

Your output must follow the format: <think>Your reasoning process</think><tool_call>Parameters</tool_call> or <think>Your reasoning process</think><answer>Your answer</answer>

Question: {question}

Options: {options}

The video lasts for {duration} seconds.

Fig. 10: The intuition prompt (S1) requiring active tool exploration.

a dedicated reflection prompt. This prompt instructs the model to critically review the reasoning behind its initial decision and revise the answer if necessary without accessing any additional visual evidence. The exact prompt used in our experiments is shown in Fig. 13.

Verification Prompt

You must ALWAYS conduct thorough reasoning inside `<think>` and `</think>` tags BEFORE answering the question. Provide your answer within `<answer>` `</answer>` tags. Your answer should be supported by evidence from the given frame. When you don't have enough visual information, please say I don't know. Your output must follow the format: `<think>`Your reasoning process`</think>``<answer>`Your answer`</answer>`
 Question: {question}
 Options: {options}
 Answer with the option's letter from the given choices within `<answer>` `</answer>` tags.

Fig. 11: The verification prompt (S2) incorporating the abstention mechanism.

Arbitration Prompt

The interaction history above presents two distinct analysis phases:
 - Phase 1 (S1): Initial Answer. (Derived from the full video and tool-retrieved frames.)
 - Phase 2 (S2): Blind Verification (based solely on retrieved frames from Initial Answer).
 Your task is to arbitrate and determine the correct answer. You must actively retrieve video frames using the tool to verify the evidence before making your final decision.
 Start your reasoning by explicitly quoting the conclusions of S1 and S2.
 You are PROHIBITED from answering directly in the first turn.
 Regardless of whether S1 and S2 agree or disagree, you MUST invoke the tool at least once to independently verify the key evidence or timestamp.
 Your output must follow the format: `<think>`...Your reasoning process...`</think>``<tool_call>`Parameters`</tool_call>` or `<think>`...`</think>``<answer>`Your answer`</answer>`
 Question: {question}
 Options: {options}
 The video lasts for {duration} seconds.
 DO NOT output `<answer>``</answer>` tags in this turn. You MUST call a tool within `<tool_call>``</tool_call>` first.

Fig. 12: The arbitration prompt (S3) designed for conflict resolution.

Reflection Prompt

Review your previous answer critically. Analyze the evidence and logic behind your initial conclusion. You might have hallucinated or missed details in the video. You must enclose your self-reflection process within `<think>` and `</think>` tags, and your final definitive answer within `<answer>` and `</answer>` tags. Your output must strictly follow this format: `<think>` Your reflection process `</think>``<answer>` Your final answer `</answer>`

Fig. 13: Reflection prompt used by the internal closed-loop reflection baseline.

F More Case Studies

To further illustrate the autonomous self-correction capabilities elicited by SD-GRPO, we present additional qualitative cases in this section.

F.1 Failure Modes of Internal Closed-Loop Reflection

We first present the failure modes of internal closed-loop reflection to further clarify the empirical observations discussed in Section 2. When objective visual evidence is absent, models that rely entirely on internal knowledge typically exhibit two dominant failure trajectories. The first trajectory manifests as the

mechanical repetition of an initially incorrect prediction (see Fig. 14). In this case, the reflection stage does not introduce new insights but instead reproduces the flawed reasoning path contained in the initial intuition. As a result, the model enters a self-reinforcing loop that consolidates the original visual hallucination. The second trajectory manifests as the erroneous overturning of a correct judgment (see Fig. 15). When the model processes complex details in long videos without external visual anchors, its internal reasoning often becomes unstable. Even when the model produces a correct answer in the initial stage, the subsequent reflection process can be misled by internally generated hallucinations, which ultimately causes the model to overturn an originally correct conclusion.

F.2 Successful Cases

Overall, these successful cases demonstrate three representative rectification trajectories. In the first pattern, the model corrects initial visual hallucinations using additionally retrieved external evidence, grounding the final arbitration strictly on objective observations (see Fig. 16). Notably, this example corresponds to the same question shown in Fig. 14, where internal closed-loop reflection fails to revise the incorrect prediction. In contrast, Reflect-R1 successfully resolves the error by retrieving external visual evidence and grounding the final decision in objective observations. In the second pattern, rather than guessing blindly when initial evidence is insufficient, the model explicitly abstains and initiates a secondary retrieval to secure a more robust conclusion (see Fig. 17). Finally, when local reasoning chains persistently conflict or fail, the model discards unreliable intermediate premises and executes a broader global search to accurately reconstruct the timeline (see Fig. 18).

F.3 Failure Cases

Although the model demonstrates robust self-correction capabilities, we observe two typical failure patterns that expose the limitations of the current architecture. The first pattern is complete failure (see Fig. 19). In such highly complex scenarios, both the initial intuition and the independent verification stage arrive at the same incorrect conclusion. Lacking a semantic conflict signal, the arbitration stage fails to initiate a deep rethinking process and directly adopts the erroneous consensus. The second pattern is arbitration failure (see Fig. 20). In these instances, the verification stage successfully yields the correct answer based on accurately retrieved frames. However, during final arbitration, the model relies excessively on the flawed initial intuition, leaving valid external evidence underutilized. These failure cases indicate that future research must focus on optimizing conflict resolution mechanisms to enhance the reasoning consistency of large models in complex multimodal tasks.

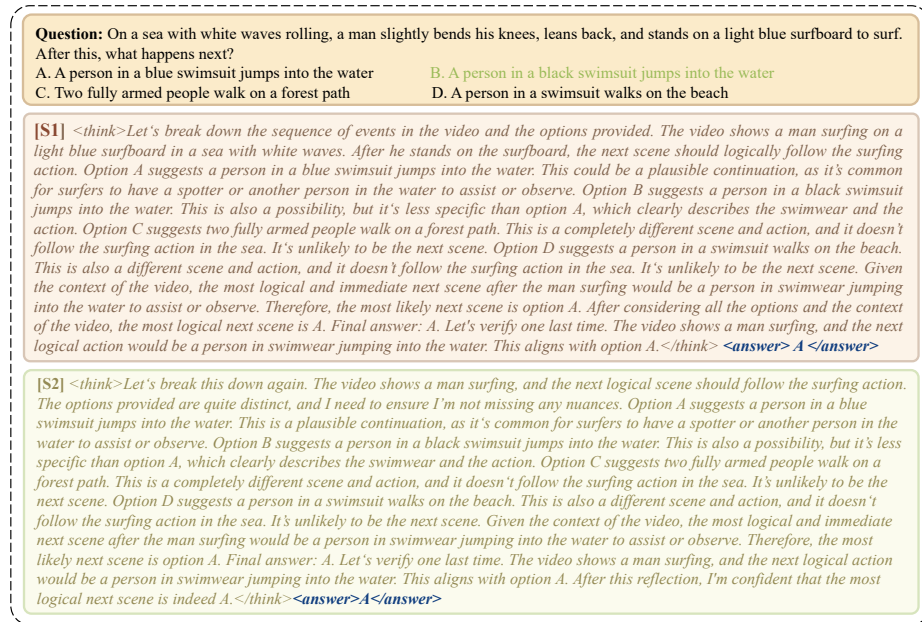


Fig. 14: Repetition of Initial Errors. During internal reflection, the model fails to generate new insights and simply repeats the erroneous reasoning of its initial intuition, trapping itself in a hallucination loop.

Question: In the upper right corner of the screen, there is a man wearing a red short-sleeved shirt. On a white background, two cups labeled with different chemical elements contain blue liquids. The liquid on the left has a paper strip marked 'Zn' in it. What happens first on the screen after the man picks up the paper strip?

- A. A pair of hands in the screen place two photos on a panel with blue liquids.
- B. Some people with badges on their necks sit facing each other at a long table outside, smiling at the camera.
- C. A woman in a brick-red suit jacket sits in front of a mirror, talking.
- D. A man in a black shirt and a woman in a white coat with a gray inner top sit in front of a mirror, and the woman's hands rest on her lap.
- E. A man in a black shirt and a woman in a white coat with a gray inner top sit in front of a mirror, and the woman opens her hands.

[S1] <think>Let's break down the sequence of events in the video. The question describes a specific scene involving a man in a red shirt and a white background with chemical elements. The man picks up a paper strip marked 'Zn'. Now, we need to determine what happens next on the screen. First, let's consider the options: A) A pair of hands in the screen place two photos on a panel with blue liquids. This option involves hands and photos, which doesn't seem to match the described scene. B) Some people with badges on their necks sit facing each other at a long table outside, smiling at the camera. This option involves people with badges and a long table, which is not related to the chemical elements. C) A woman in a brick-red suit jacket sits in front of a mirror, talking. This option involves a woman in a brick-red suit, which is not related to the chemical elements. D) A man in a black shirt and a woman in a white coat with a gray inner top sit in front of a mirror, and the woman's hands rest on her lap. This option involves a man and a woman in a white coat, which is not related to the chemical elements. E) A man in a black shirt and a woman in a white coat with a gray inner top sit in front of a mirror, and the woman opens her hands. This option involves a man and a woman in a white coat, and the woman opens her hands, which is a plausible next action after picking up the paper strip. Given the context of the chemical elements and the man picking up the paper strip, the most logical next action is for the woman to open her hands, as it is a common gesture after handling a paper strip. Therefore, the correct answer is E. </think> <answer>A </answer>

[S2] <think>Let's break down the sequence of events in the video and the options provided. The question describes a specific scene involving a man in a red shirt and a white background with chemical elements. The man picks up a paper strip marked 'Zn'. Now, we need to determine what happens next on the screen. First, let's consider the options: A) A pair of hands in the screen place two photos on a panel with blue liquids. This option involves hands and photos, which doesn't seem to match the described scene. B) Some people with badges on their necks sit facing each other at a long table outside, smiling at the camera. This option involves people with badges and a long table, which is not related to the chemical elements. C) A woman in a brick-red suit jacket sits in front of a mirror, talking. This option involves a woman in a brick-red suit, which is not related to the chemical elements. D) A man in a black shirt and a woman in a white coat with a gray inner top sit in front of a mirror, and the woman's hands rest on her lap. This option involves a man and a woman in a white coat, which is not related to the chemical elements. E) A man in a black shirt and a woman in a white coat with a gray inner top sit in front of a mirror, and the woman opens her hands. This option involves a man and a woman in a white coat, and the woman opens her hands, which is a plausible next action after picking up the paper strip. Given the context of the chemical elements and the man picking up the paper strip, the most logical next action is for the woman to open her hands, as it is a common gesture after handling a paper strip. Therefore, the correct answer is E. </think> <answer>E </answer>

Fig. 15: Overturning a Correct Initial Answer during Internal Reflection. Without external objective anchors, the model becomes misled by internally generated hallucinations during reflection, mistakenly overturning its initially correct answer.

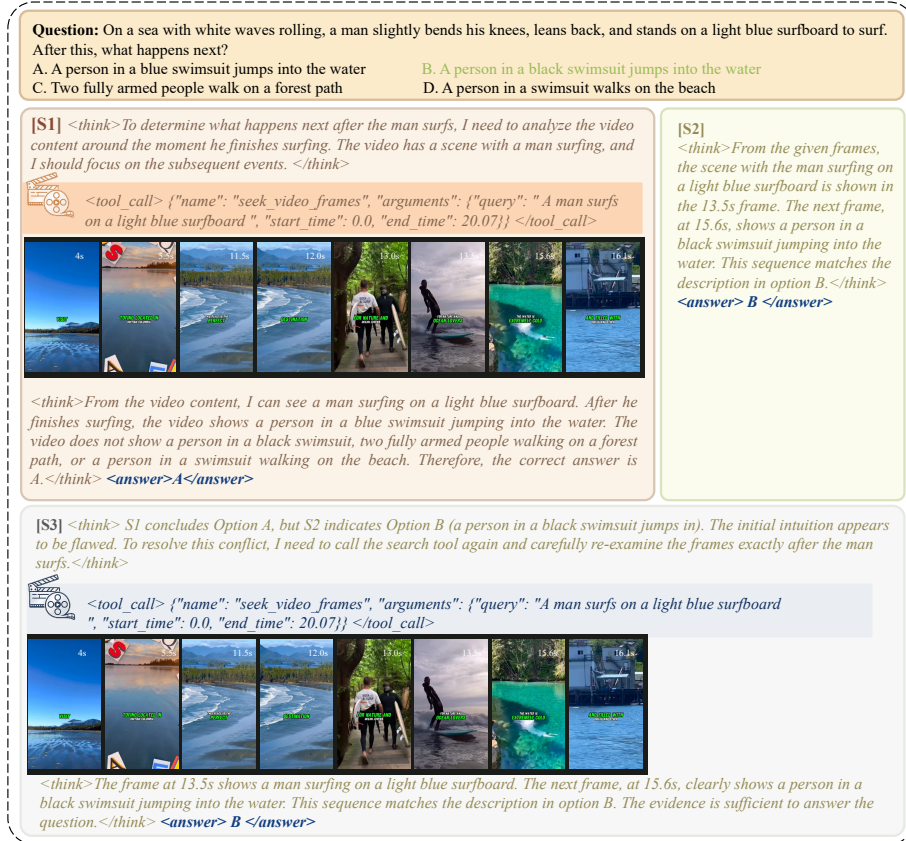


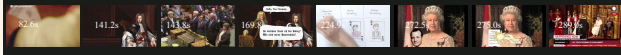
Fig. 16: Evidence-Grounded Hallucination Rectification. The model resolves the initial visual hallucination by retrieving and grounding its final arbitration strictly on external objective frames.

Question: The white-robed lady wearing the crown sits on a golden chair, with golden ornaments hanging on her chest. On both sides of her are two men, each with various badges on their outer clothes. One man with a golden cape is bending in front of the lady. What did the crowned lady do on her first appearance?

- A. Accepted the item handed over by the man on the right B. Accepted the item handed over by the man on the left
C. Refused the item handed over by the man on the right D. Accepted the item handed over by the man who was bending

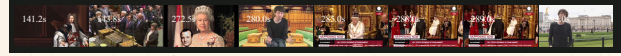
[S1] <think>From the video frames, I can see a crowned lady sitting on a golden chair with golden ornaments. There are two men on either side of her, and one of them is bending in front of her. However, the frames do not show the interaction between the crowned lady and the man who is bending. To determine what the crowned lady did on her first appearance, I need to analyze the video content related to her initial scene.</think>

<tool_call> {"name": "seek_video_frames", "arguments": {"query": "The white-robed lady wearing the crown sits on a golden chair, with golden ornaments hanging on her chest. On both sides of her are two men, each with various badges on their outer clothes. One man with a golden cape is bending in front of the lady.", "start_time": 0.0, "end_time": 341.6}} </tool_call>



<think>From the video frames, I can see a crowned lady sitting on a golden chair with golden ornaments. There are two men on either side of her, and one of them is bending in front of her. However, the frames do not show the interaction between the crowned lady and the man who is bending. To determine what the crowned lady did on her first appearance, I need to analyze the video content related to her initial scene.</think>

<tool_call> {"name": "seek_video_frames", "arguments": {"query": "The crowned lady sitting on a golden chair with golden ornaments, and the man bending in front of her.", "start_time": 0.0, "end_time": 341.6}} </tool_call>



<think>From the video frames, I can see a crowned lady sitting on a golden chair with golden ornaments. There are two men on either side of her, and one of them is bending in front of her. The crowned lady is wearing a crown and has golden ornaments on her chest. The man bending in front of her is wearing a golden cape. The frames show the interaction between the crowned lady and the man who is bending. The crowned lady is accepting the item handed over by the man who is bending. Therefore, the crowned lady accepted the item handed over by the man who was bending.</think> <answer>D</answer>

[S2] <think>From the video frames, the crowned lady is seated on a golden chair with golden ornaments. On both sides of her are two men, each with various badges on their outer clothes. One man with a golden cape is bending in front of the lady. The video frames do not show any interaction between the crowned lady and the man bending, nor do they show any items being handed over. Therefore, there is no visual evidence to determine what the crowned lady did on her first appearance.</think>

<answer>I don't know</answer>

[S3] <think>Extraction: S1=D S2=I don't know My initial analysis from Phase 1 and Phase 2 suggests a discrepancy. To ensure accuracy, I need to verify the key evidence from the video frames related to the crowned lady's first appearance.</think>

<tool_call> {"name": "seek_video_frames", "arguments": {"query": "The white-robed lady wearing the crown sits on a golden chair, with golden ornaments hanging on her chest. On both sides of her are two men, each with various badges on their outer clothes. One man with a golden cape is bending in front of the lady.", "start_time": 0.0, "end_time": 341.6}} </tool_call>



<think>From the video frames, I can see a crowned lady sitting on a golden chair with golden ornaments. There are two men on either side of her, and one of them is bending in front of her. The crowned lady is wearing a crown and has golden ornaments on her chest. The man bending in front of her is wearing a golden cape. The frames show the interaction between the crowned lady and the man who is bending. The crowned lady is accepting the item handed over by the man who is bending. Therefore, the crowned lady accepted the item handed over by the man who was bending.</think> <answer>D</answer>

Fig. 17: Abstention and Re-retrieval under Missing Evidence. When initial retrieval provides insufficient evidence, the model explicitly abstains from guessing and executes a secondary search to secure robust verification.



Fig. 18: Global Reconstruction after Local Reasoning Failure. When prior multi-stage reasoning yields conflicting and incorrect sequences, the arbitrator discards local premises and initiates a global video search to reconstruct the correct timeline.

Question: Which of the following sequences of scenes is correct?

A. First, the camera moves to the right, revealing a deep red area on the map labeled Valley. Then, the camera slowly zooms in, showing a map with a light pink area labeled Delaware Valley. Finally, two markers appear on the map pointing to two locations.

B. First, the camera slowly zooms in, showing a map with a light pink area labeled Delaware Valley. Then, the camera moves to the right, revealing a deep red area on the map labeled Valley. Finally, two markers appear on the map pointing to two locations.

C. First, the camera slowly zooms in, showing a map with a light pink area labeled Delaware Valley. Then, two markers appear on the map pointing to two locations. Finally, the camera moves to the right, revealing a deep red area on the map labeled Valley.

D. First, two markers appear on the map pointing to two locations. Then, the camera slowly zooms in, showing a map with a light pink area labeled Delaware Valley. Finally, the camera moves to the right, revealing a deep red area on the map labeled Valley.

[S1] <think>To determine the correct sequence of scenes, I need to analyze the video content. The video shows a map with various labeled areas and markers. I should use the 'seek_video_frames' tool to observe the sequence of events in the video.</think>

 <tool_call> {"name": "seek_video_frames", "arguments": {"query": "The camera moves to the right, revealing a deep red area on the map labeled Valley.", "start_time": 0.0, "end_time": 17.02}} </tool_call>



<think>From the video content, the sequence of scenes is as follows: 1. The camera slowly zooms in, showing a map with a light pink area labeled Delaware Valley. 2. The camera moves to the right, revealing a deep red area on the map labeled Valley. 3. Two markers appear on the map pointing to two locations. This matches option B. The sequence of scenes is correct.</think> <answer>B </answer>

[S2] <think>From the sequence of frames, it is observed that the camera initially shows a map with a light pink area labeled Delaware Valley. Then, the camera slowly zooms in, and two markers appear on the map pointing to two locations. Finally, the camera moves to the right, revealing a deep red area on the map labeled Valley. This sequence matches option B. </think>

<answer>B </answer>

[S3] <think>S1=B, S2=B. Both Phase 1 and Phase 2 analysis concluded that the correct sequence is Option B. However, to ensure accuracy, I need to independently verify the key evidence from the video content.</think>

 <tool_call> {"name": "seek_video_frames", "arguments": {"query": "The camera slowly zooms in, showing a map with a light pink area labeled Delaware Valley.", "start_time": 0.0, "end_time": 17.02}} </tool_call>



<think>From the video content, the sequence of scenes is as follows: 1. The camera slowly zooms in, showing a map with a light pink area labeled Delaware Valley. 2. The camera moves to the right, revealing a deep red area on the map labeled Valley. 3. Two markers appear on the map pointing to two locations. This matches option B. The sequence of scenes is correct.</think> <answer>B </answer>

Fig. 19: Complete Failure. Both the intuition and verification stages arrive at the same incorrect conclusion. Lacking a conflict signal, the arbitrator adopts the shared error instead of initiating further exploration.

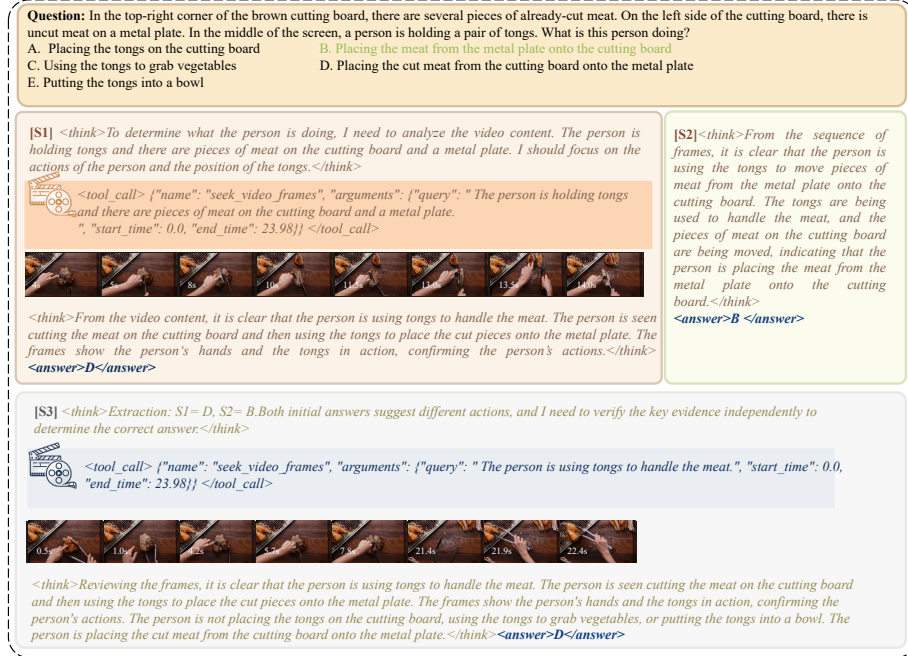


Fig. 20: Arbitration Failure. The verification stage successfully identifies the correct answer, but the arbitration stage erroneously reverts to the flawed initial intuition.